Memory Protection

Lecture 8

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Memory Protection

- Memory Protection Unit
  - Privilege Modes
  - Regions
- Memory Management Unit
  - Pages
  - Frames
  - TLB

Memory Protection

ARM: MPU, RISC-V: PMP

MPU for RP2040

Protected Memory System Architecture v7 (PMSAv7)

Bibliography

for this section

Joseph Yiu, *The Definitive Guide to ARM® Cortex®-M0 and Cortex-M0+ Processors, 2nd Edition*

- Chapter 12 - *Memory Protection Unit*

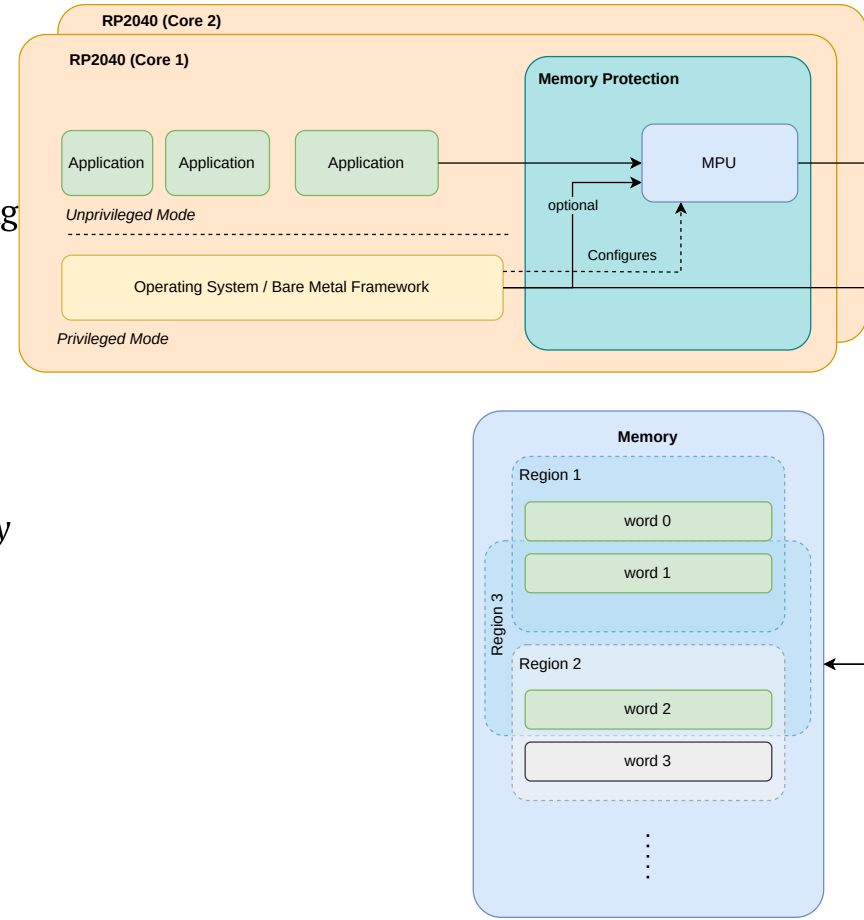
MPU for RP2040

Cortex-M0+ works in three modes

- **handler** mode - *no restrictions* - used while executing ISRs and Exception Handlers
- **thread** mode
  - **privileged** *no restrictions* - usually used for the operating system
  - **unprivileged** mode - *allows only ALU and memory access through Memory Protection* - used for applications

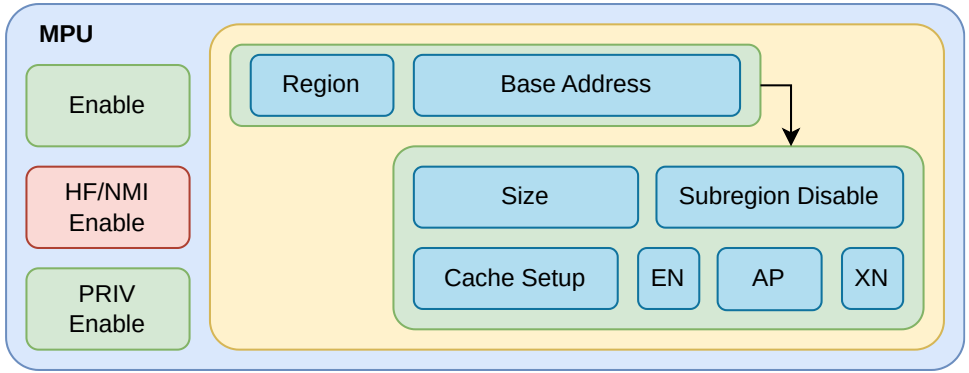
MPU allows 8 regions

- each region has up to 8 subregions
- permissions R W X



Memory Protection Unit

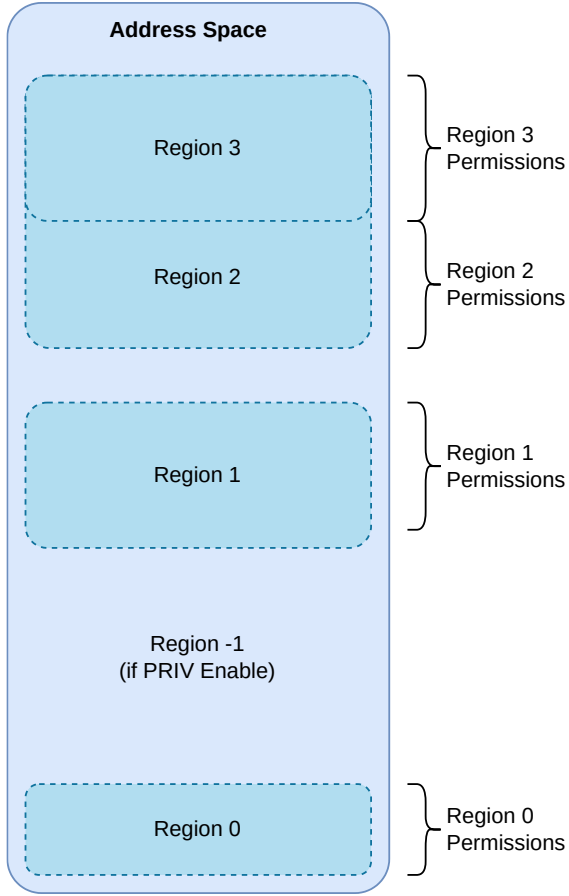
Cortex-M MPU (PMSAv7-m)



- allows the definition of *memory regions*
- regions can overlap, *highest region number* takes *priority*
- regions have access permissions (similar to rwx)

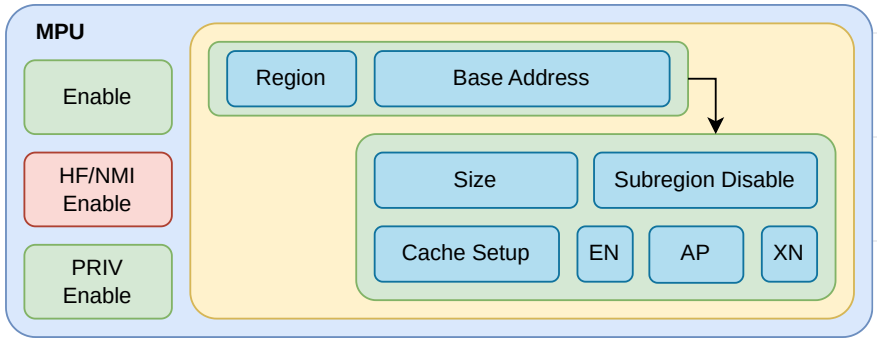
$$region_size = 2^{size}, size >= 8$$

$$base_address = region_size \times N$$



Memory Protection Unit

Access Protection



AP Access Protection

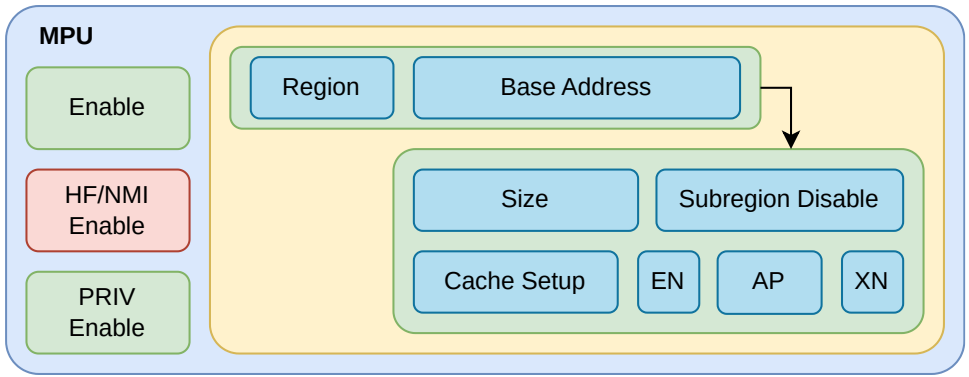
XN eXecute Never

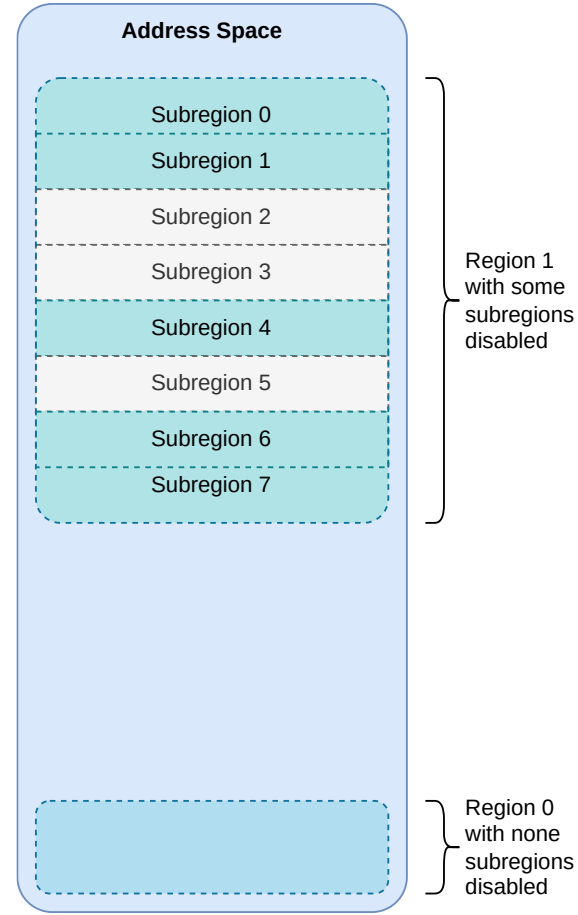
- faults if MCU has to read the next instruction from an *XN* region

AP	Privileged Mode	Unprivileged Mode
`000`	No Access	No Access
`001`	Read/Write	No Access
`010`	Read/Write	Read only
`011`	Read/Write	Read/Write
`100`	Do not use	Do not use
`101`	Read only	No Access
`110`	Read only	Read only
`111`	Read/Write	Read only

Subregions

- each region is divided in 8 subregion
- each bit in `Subregion Disable` disables a subregion
- a disabled subregion triggers a fault if accessed





Subregions' Usage

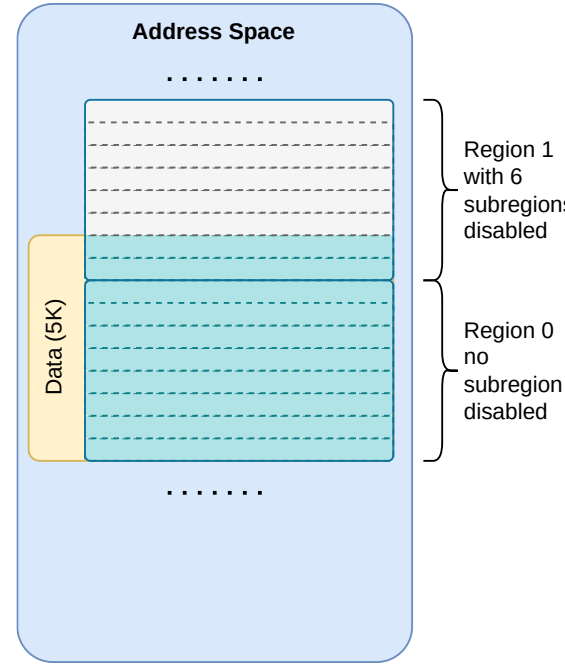
improve granularity

$$region_size = 2^{size}, size >= 8$$

$$base_address = region_size \times N$$

$$subregion_size = \frac{region_size}{8}$$

- a 5K region is not allowed (5K is not a power of 2)
- use two 4K regions back to back
- disable 6 of the subregions (subregion is 512B)



Memory Layout

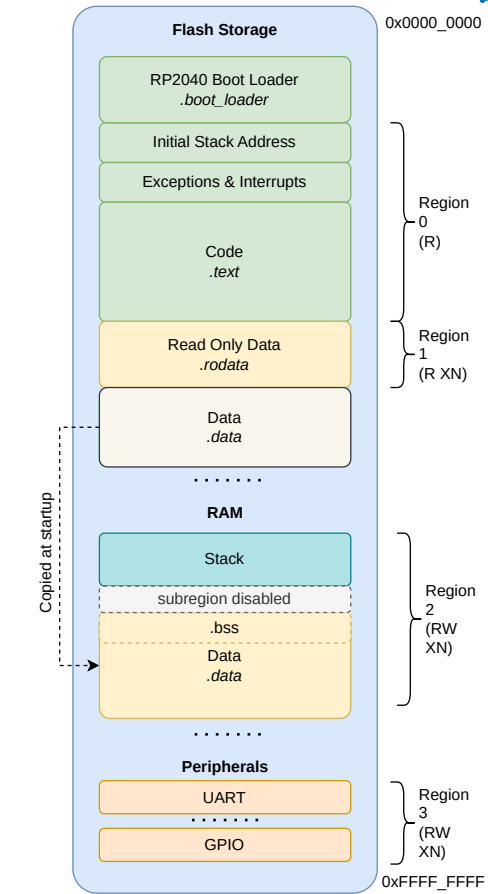
protection

Flash

- **Code** - *read* and *execute*
- **.rodata** - constants - *read only*
- **.data** - *in flash* - initialized global variables
  - is copied to RAM at startup by the `init` function
  - *should not be accessed after startup*

RAM

- **stack** - *read* and *write*
  - *usually protected by unaccessible memory before and after*
- **.data** - *in RAM* - global variables - *read* and write
- **.bss** - global variables (not initialized or initialized to `0`) - *read* and *write*



* drawing is not at scale, code and data are significantly greater than the interrupt vector

MPU for RP2350

Protected Memory System Architecture v8 (PMSAv8)

Bibliography

for this section

Joseph Yiu, *The Definitive Guide to ARM® Cortex®-M23 and Cortex-M33 Processors*

- Chapter 6 - *Memory System*
  - Subchapter 6.4 - *Access Permission Management*
- Chapter 12 - *Memory Protection Unit (MPU)*

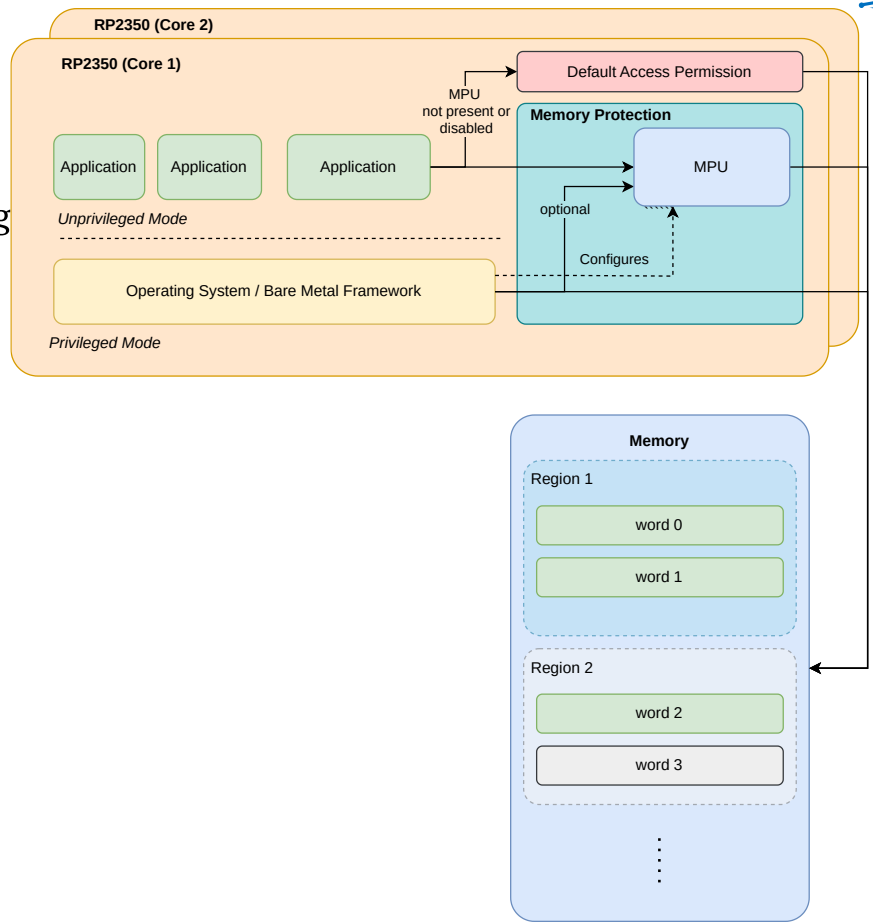
MPU for RP2350

Cortex-M33 works in three modes

- **handler** mode - *no restrictions* - used while executing ISRs and Exception Handlers
- **thread** mode
  - **privileged** *no restrictions* - usually used for the operating system
  - **unprivileged** mode - *used for applications*, allows only ALU and memory access through:
    - Default Access Permission - *restricts unprivileged access to the Cortex-M Peripherals*
    - Memory Protection

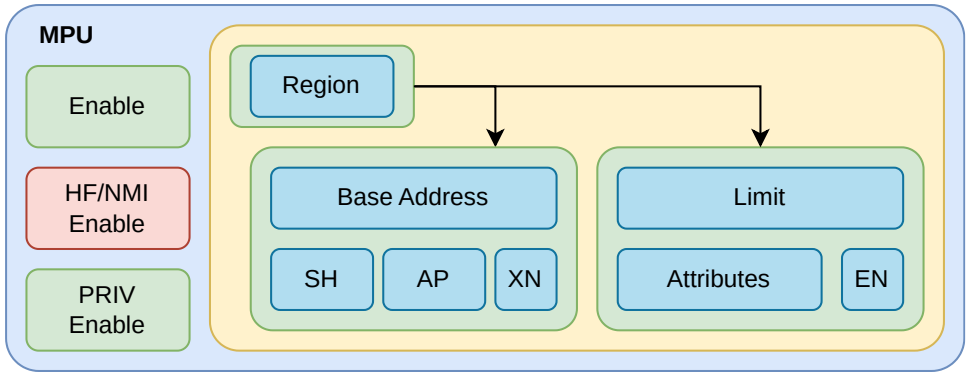
MPU allows 8 regions

- permissions R W X



Memory Protection Unit

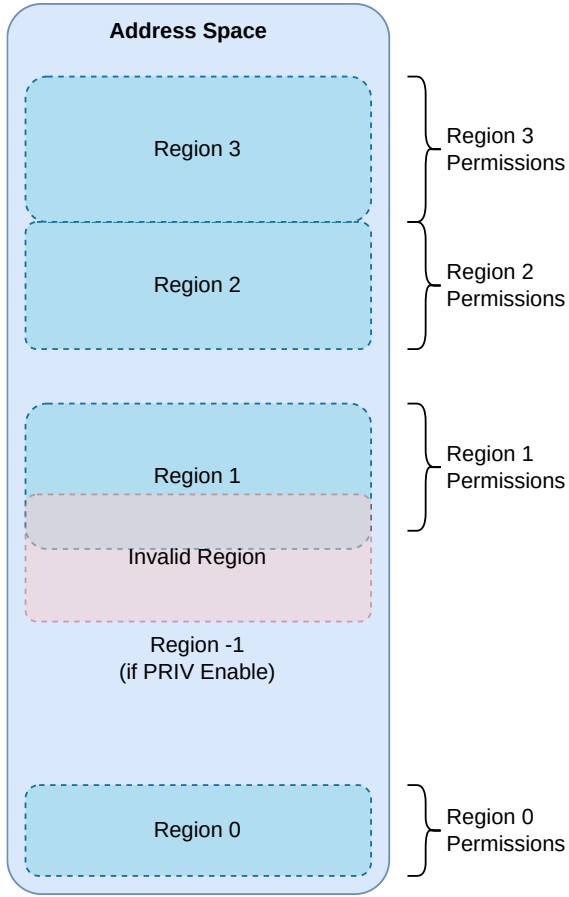
Cortex-M MPU (PMSAv8)



- allows the definition of *memory regions*
- regions cannot overlap
- regions have access permissions (similar to rwx)

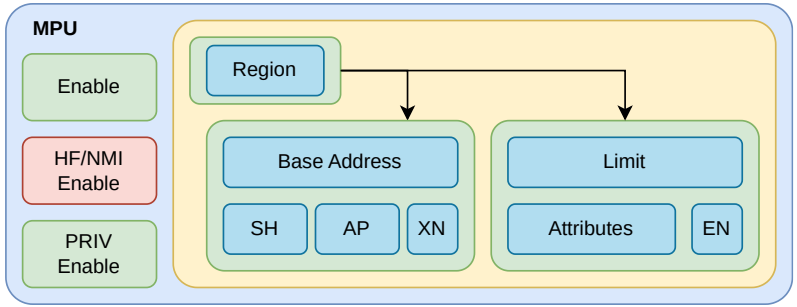
$$region_size = 32 \times N$$

$$base_address = 32 \times N$$



Memory Protection Unit

Access Protection



AP Access Protection

XN eXecute Never

- faults if MCU has to read the next instruction from an *XN* region

SH Shared between cores and peripherals

Attributes *used for cache*

AP	Privileged Mode	Unprivileged Mode
`00`	Read/Write	No Access
`01`	Read/Write	Read/Write
`10`	Read only	No Access
`11`	Read only	Read only

Better granularity -> there is no need for *No Access* in privileged mode.

> There is no need to overlap regions to obtain the required protected memory space.

Memory Management

MMU

Bibliography

for this section

1. **Andrew Tanenbaum**, *Modern Operating Systems (4th edition)*
   - Chapter 3 - *Memory Management*
     - Subchapter 3.3 - *Virtual Memory*
2. **Philipp Oppermann**, *Writing an OS in Rust*
   - *Introduction to Paging*
   - *Paging Implementation*

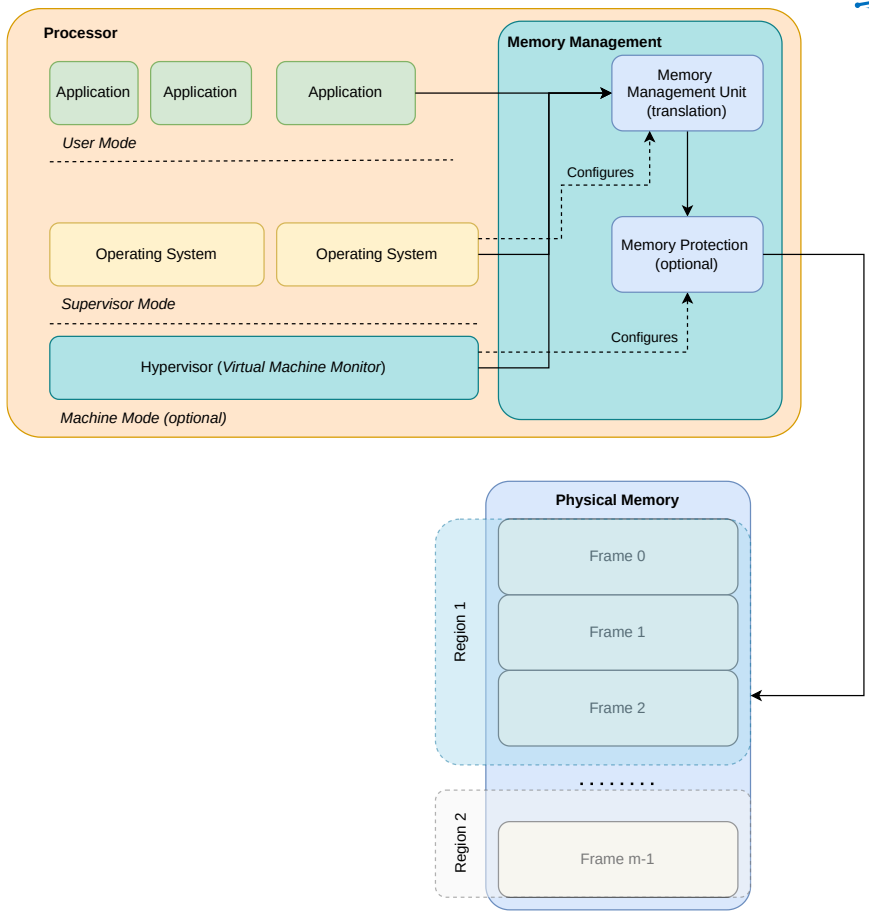
Memory Management

memory access defined page by page

- uses *logical addresses*
- **translates** to *physical addresses*

The processor works in at least two modes:

- **supervisor** mode
  - restricts access to some registers
  - accesses virtual addresses through Memory Protection (*if machine mode exists*)
- **user** mode
  - allows only ALU and memory load and store
  - accesses memory access through the Memory Management Unit (*MMU*)



Paging

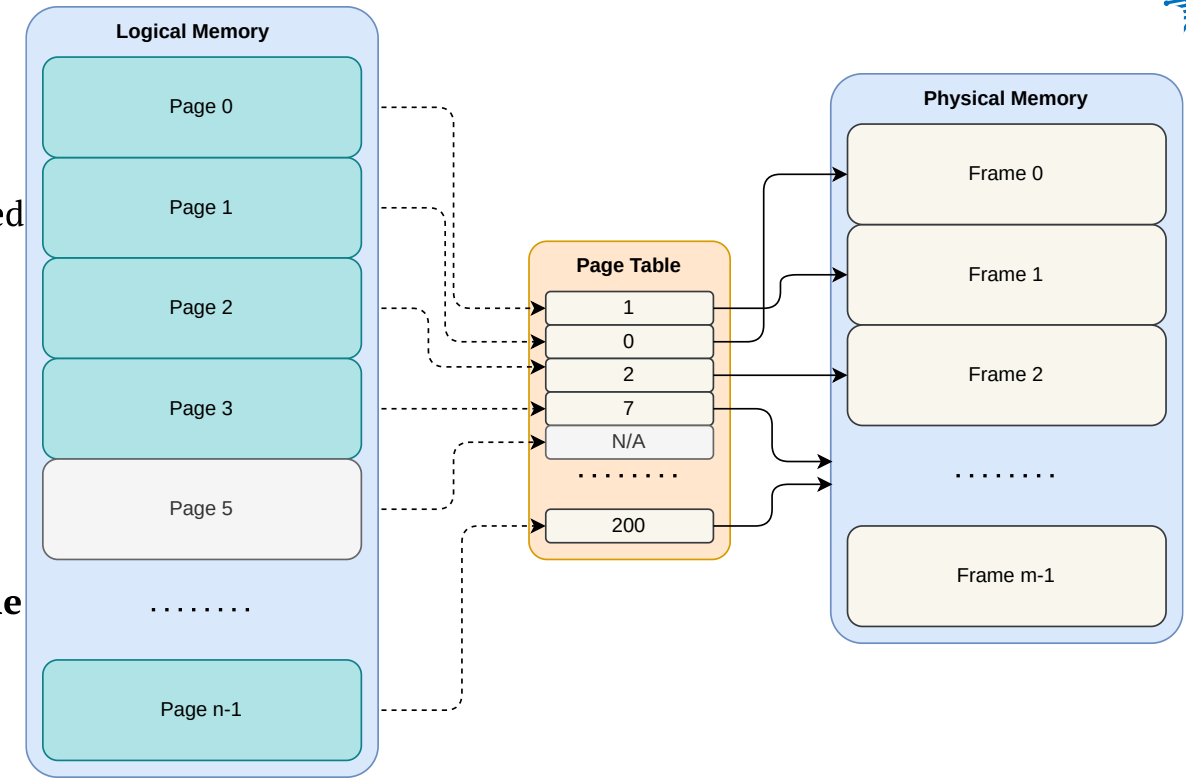
the memory *unit* is the page

- Physical Memory (*RAM*) is divided in **frames**
- Logical Memory is divided in **pages**
- *page* = *frame* = **4 KB** (usually)

logical addresses are translated to *physical addresses* using a **page table**

the **page table** is located in the **physical memory**

- each memory access requires at least 2 memory accesses



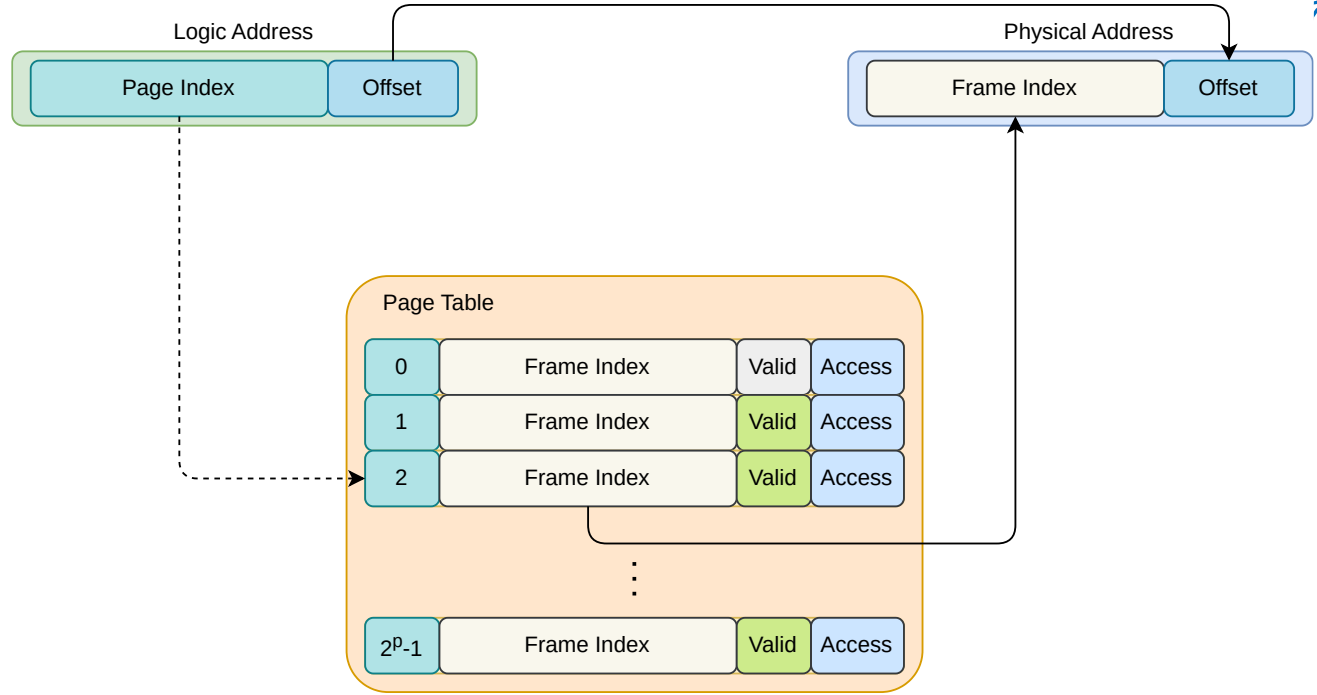
Address Translation

page to frame

the logic address is divided in two parts:

- *page index*
- *offset* within the page

the MMU translates every logic address into a physical address using a *page table*



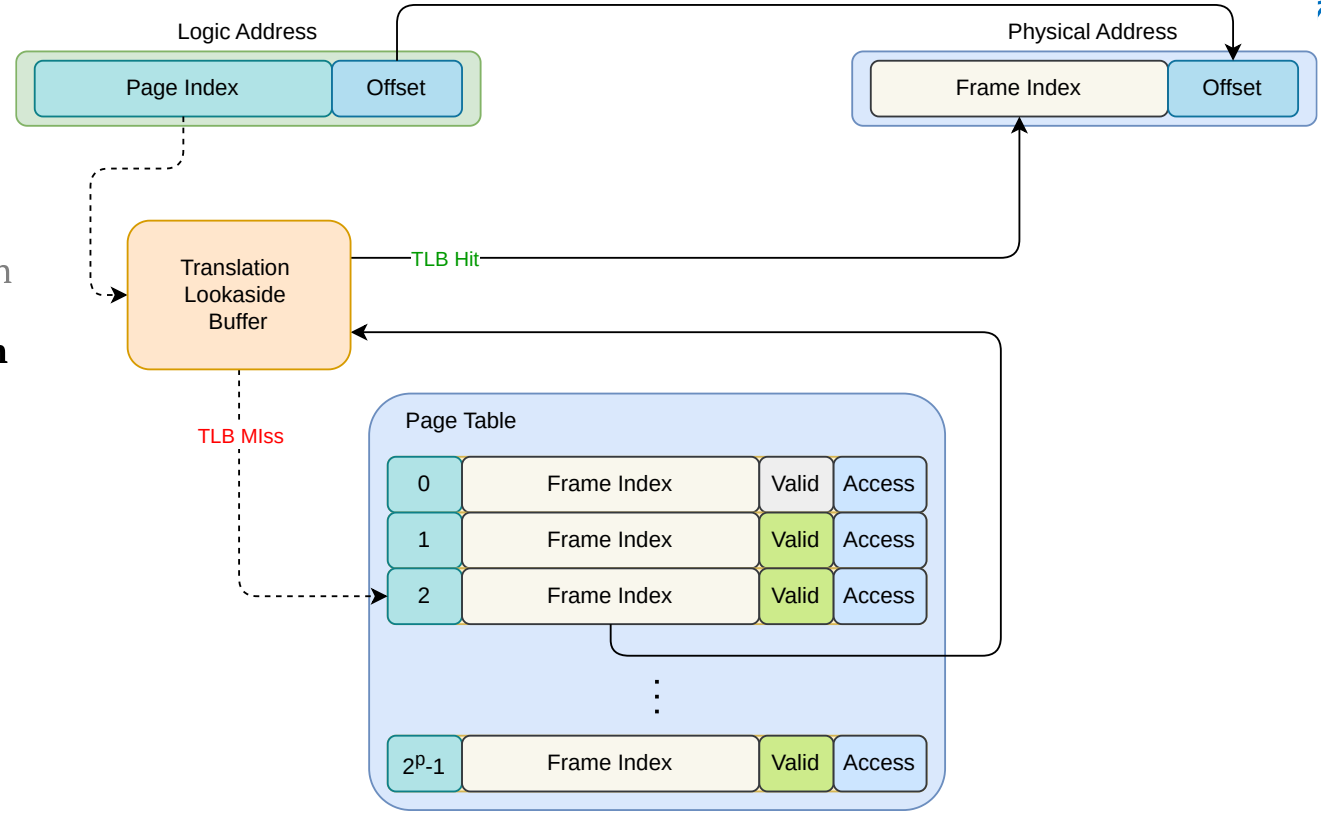
Translation Lookaside Buffer (TLB)

caching address translation

the **page table** is **stored in RAM**

each memory access **requires 2 accesses**

1. read the page table entry to translate the address
2. the requested access



Page Directory

caching address translation

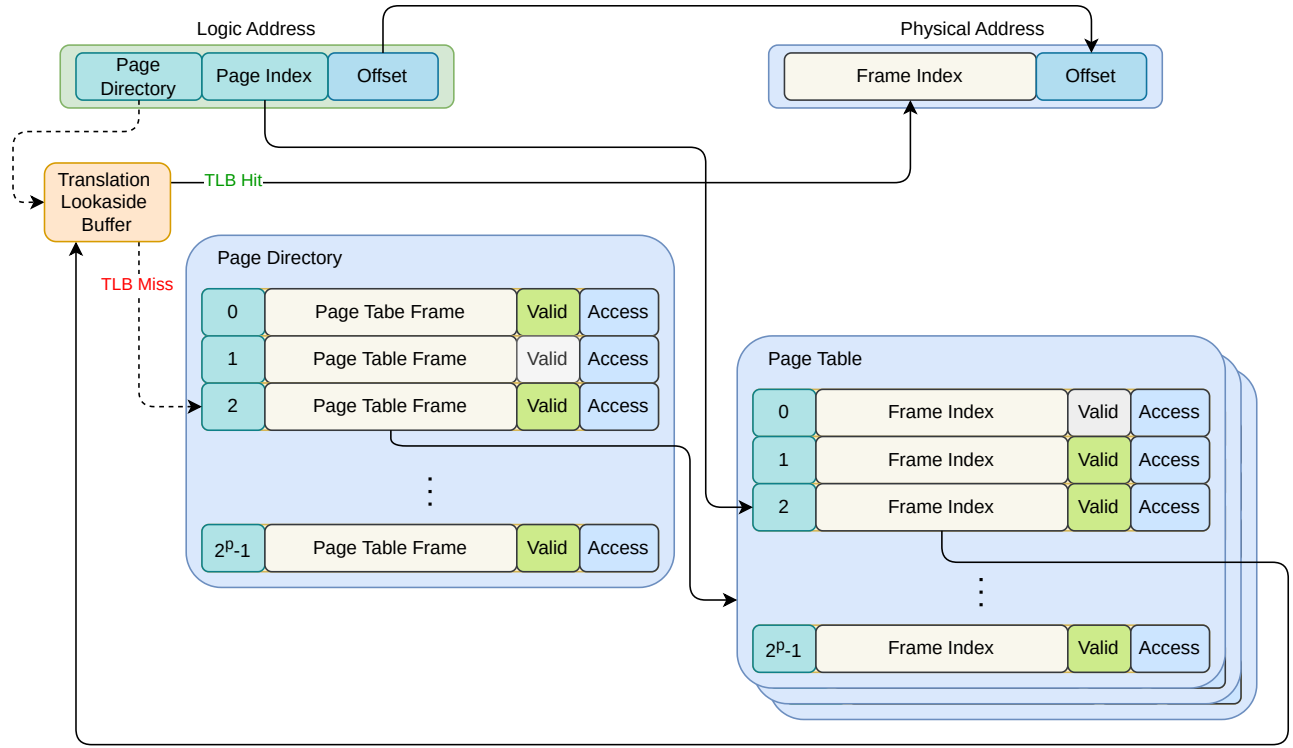
$$size_{table} = \frac{size_{ram}}{size_{page}}$$

- each table entry is 4B
- the address space is 4GB (for 32 bits processors)

$$size_{table_32_bits} = \frac{2^{32}}{4 \times 2^{10}}$$

$$size_{table_32_bits} = 4MB$$

RAM was counted in MB when paging started being used



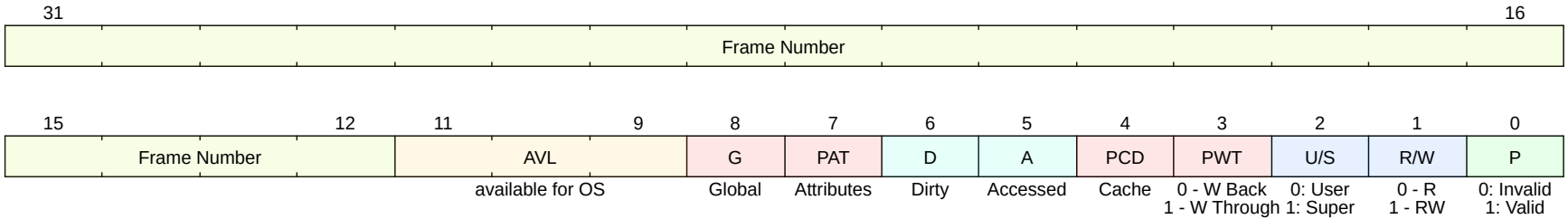
two levels, page directory and table, usually used for 32 bits systems

Page Table Entry

for x86 - 32 bits

this is one entry of the page table

- **P** - is the page's frame present in RAM?
- **R/W** - read only or read write access
- **U/S** - can the page be accessed in user mode?
- **D** and **A** - has this page been written since the OS has reset these bits?
- **AVL** - bits available for the OS to use, ignored by MMU

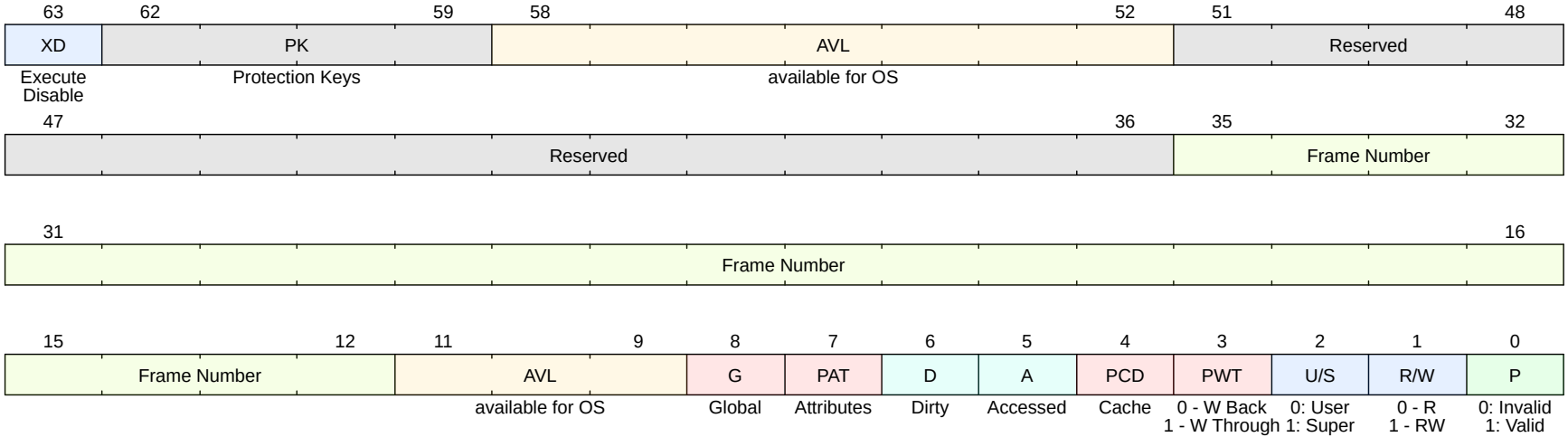


Page Table Entry

for x86 - 32 bits with PAE

this is one entry of the page table using Physical Address Extension (*PAE*)

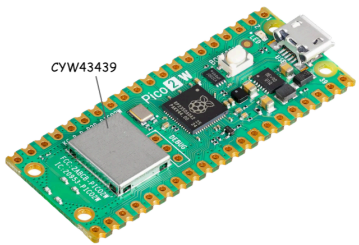
- **XD** - eXecute Disable (aka *DEP*), if set triggers a fault if an instruction is read from the page
- **PK** - Protection Keys, allows user mode to set protection (64 bit only)



Microcontroller (MCU)

Integrated in embedded systems for certain tasks

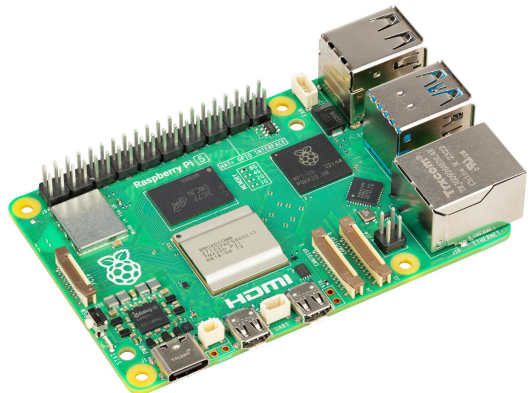
- low operating frequency (MHz)
- a lot of I/O ports
- controls hardware
- does not require an Operating System
- costs $0.1 - $25
- uses **Memory Protection Unit**



Microprocessor (CPU)

General purpose, for PC & workstations

- high operating frequency (GHz)
- limited number of I/O ports
- usually requires an Operating System
- costs $75 - $500
- uses **Memory Management Unit**



Conclusion

we talked about

- Memory Protection Unit
  - Privilege Modes
  - Regions
- Memory Management Unit
  - Pages
  - Frames
  - TLB